The Role of Generative AI Chatbots in Higher Education: a Student-Centric Conceptual Analysis of Benefits, Ethics, and Privacy Concerns

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Abstract

Generative AI chatbots, such as ChatGPT, are reshaping higher education by offering personalized tutoring, administrative support, and enhanced research capabilities, while simultaneously raising ethical and privacy concerns. This study examines their role from a student-centered perspective, focusing on benefits, risks, and institutional implications. A pilot study was carried out with 300 undergraduate students and 120 faculty members from five European universities. Participants engaged in semi-structured interviews, perception surveys, and controlled academic tasks designed to evaluate chatbot performance. Instruments included standardized student satisfaction questionnaires, expert review rubrics, and quantitative performance metrics (perplexity, response latency, context window, SWE-bench accuracy). Data were analyzed through descriptive statistics, ANOVA tests, and thematic coding of interviews. Results reveal that ChatGPT and Claude achieved the best balance between pedagogical clarity and privacy compliance, while Gemini excelled in technical capacity but showed weaker ethical safeguards. Student satisfaction was strongly associated with transparency in data policies and pedagogical usefulness rather than raw technical performance. These findings highlight the need for universities to adopt

generative AI technologies under robust ethical and privacy frameworks. Future research should extend this pilot into longitudinal studies and institutional case analyses.